

Methods	Year	Parameters	Dice	iou	Input size
TransNetR (4)	2023	27.27	87.06	80.16	?
SegMed-Net (8)	2026	59.9	90.8	83.70	352*352
PVT-Cascade (3)	2023	35.2	91.1	86.3	352*352
MegaNet (2)	2024	44.1	91.3	86.3	352*352
Fu-TransHNet (5)	2023	?	91.6	86.6	352*352
LiteBoundD (1)	2026	32.8	91.9	84.1	352*352
AAPCNet (7)	2025	38.2	92.10	86.9	256*256
DVFIP-Net	2026	45.62	92.20	87.40	352*352
SAM2-FNet	2026	27.2	92.40	87.40	352*352
MLB-Net	2026	87.7	92.60	87.80	352*352
MSPN	2026	28.56	92.70	87.90	352*352
HCA-Net	2026	89.70	93.60	89.7	352*352
Ours	2026	38.5	93.89	88.66	352*352
PolyMamba-Net (6)	2026	25.3	94.2	89.1	352*352

1. Sharpening Lightweight Models for Generalized Polyp Segmentation: A Boundary Guided Distillation from Foundation Models
2. “Meganet: Multi-scale edge-guided attention network for weak boundary polyp segmentation
3. “Medical image segmentation via cascaded attention decoding
4. TransNetR: Transformer-based Residual Network for Polyp Segmentation with Multi-Center Out-of-Distribution Testin
5. Cooperation Learning Enhanced Colonic Polyp Segmentation Based on TransformerCNN Fusion
6. PolyMamba-Net: a lightweight and boundary-aware network for real-time polyp segmentation in colonoscopy
7. Attention-Guided Asymmetric Multiscale Polyp Segmentation Network
8. SegMed-Net: a domain-aware dual encoder-decoder network with dense feature fusion for enhanced medical image segmentation

Boundary-Frequency Injection Module:

- Decomposing frequency components
 - Frequency Feature Decomposer (FFD)
 - Spilt skip features into high-frequency and low-frequency using 2D FFT
 - * High-frequency -> Boundaries, edges
 - * Low-frequency -> semantic content
- Using boundary information for better feature guidance
 - Boundary-guided Squeeze-Excitation Injection (BSEI)
 - Uses Sobel operators to detect boundaries in skip features
 - Creates spatial attention maps from boundary information
 - Uses cross-attention between decoder features and boundary-weighted encoder features
- Adaptively fusing these different feature representations
 - Multi-scale Frequency Adaptive Fusion (MFAF)
 - Learns soft weights to combine high-frequency, low-frequency, and boundary-enhanced features

- Uses channel-wise gating to decide the importance of each component

Purpose:

The BFIM replaces standard skip connections to:

- **Better preserve boundary information** during up-sampling
- **Adaptively emphasize** important frequency components
- **Improve segmentation accuracy** especially for object boundaries
- **Provide learnable fusion** instead of simple concatenation

Models	Seen				Unseen						Epoch
	Kvasir-Seg		ClinicDB		CVC-300		ColonDB		ETIS		
	Dice	iou	Dice	iou	Dice	iou	Dice	iou	Dice	iou	
HCA-Net	93.6	89.7	95.6	90.9	91.2	87.3	83.4	76.9	83.3	78.9	
Ours	92.40	86.08	95.19	90.84	91.49	84.32	81.26	70.44	87.39	77.61	1370
Ours-2	92.58	86.41	95.31	91.04			81.45	70.95			1405
	92.03		95.27	90.97			83.34	73.75			

Models	Seen				Unseen						Epoch
	Kvasir-Seg		ClinicDB		CVC-300		ColonDB		ETIS		
	Dice	iou	Dice	iou	Dice	iou	Dice	iou	Dice	iou	
HCA-Net	93.6	89.7	95.6	90.9	91.2	87.3	83.4	76.9	83.3	78.9	
	93.49	87.77	95.31	91.04	91.61	84.52	83.34	73.75	88.69	79.67	
Our	93.49	87.77	94.92	90.33	90.33	82.52	79.72	69.16	88.69	79.67	100
Ours	92.58	86.41	95.31	91.04	91.49	84.32	83.34	73.75	87.39	77.61	1370
Ours-93.04	92.30				91.25	83.91			84.96	73.85	
Ours-93.66	92.75	86.48	95.20	90.83	91.61	84.52	82.81	72.96	85.39	74.50	2836

Introduction

Colorectal polyps are abnormal tissue growths that can develop into malignant lesions if they are not detected and removed at an early stage. Colonoscopy is widely used for the examination of the lower gastrointestinal tract and enables clinicians to identify and remove suspicious lesions. Nevertheless, visual inspection remains challenging because polyp appearance varies considerably across patients and imaging conditions. Variations in lesion size, morphology, texture, contrast, and location, together with specular reflections, motion blur, intestinal folds, bubbles, and other visual artifacts, can complicate accurate delineation of polyp regions. Automated polyp segmentation therefore remains an important problem in computer-aided colonoscopy, where reliable pixel-level localization may assist subsequent analysis and clinical decision support.

Accurate polyp segmentation requires a model to simultaneously capture high-level semantic information and preserve sufficiently detailed spatial representations. U-shaped encoder-decoder architectures are well suited to this task because they combine progressively abstract encoder features with higher-resolution representations through skip connections. U-Net established this general design, while subsequent architectures such as UNet++ further demonstrated the importance of multi-level feature interaction and hierarchical decoding [1, 2]. However, repeated downsampling reduces spatial resolution, and fixed decoder operations

may not always provide sufficient flexibility for lesions that vary considerably in scale and visual frequency content.

Recent transformer-based and hybrid architectures have improved long-range dependency modeling in medical image segmentation. TransUNet combines convolutional representations with transformer encoding, while Polyp-PVT applies hierarchical transformer features and cross-level aggregation to polyp segmentation [3, 4]. These approaches demonstrate the importance of broad contextual modeling, but they can introduce additional architectural complexity. Modern convolutional networks such as ConvNeXt provide an alternative by retaining the efficiency and spatial inductive bias of convolution while incorporating design principles that improve representation capacity. ConvNeXt employs large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth, making it a strong hierarchical encoder for dense prediction tasks [5].

Multi-scale representation is particularly important in polyp segmentation because lesions may occupy very small regions or extend over a large portion of the image. Multi-Scale Convolution Blocks (MSCBs) address this issue by combining convolutional branches with different receptive fields, allowing a network to aggregate local and broader spatial responses. However, conventional multi-branch aggregation generally combines the branch responses using a fixed formulation that is shared across all input images. Consequently, the relative contribution of different receptive fields is not explicitly conditioned on the frequency characteristics of each individual sample.

Frequency-aware feature processing provides another complementary perspective. Frequency-Aware Feature Enhancement Modules (FAFEMs) decompose deep representations into low- and high-frequency components and use these components to enhance feature responses. Low-frequency representations mainly preserve broader semantic structures, whereas high-frequency representations emphasize local spatial variations and fine structural information. Existing frequency-aware modules are typically used to improve the feature representation at the location where the frequency decomposition is performed. However, the frequency descriptor extracted from deep semantic representations can potentially provide useful global information for guiding later multi-scale processing.

This observation motivates the central idea of this work. Instead of treating bottleneck frequency enhancement and decoder multi-scale refinement as independent operations, we establish an explicit cross-level interaction between them. The frequency descriptor produced by a bottleneck FAFEM is propagated to the first decoder stage and used to regulate the relative contributions of the multi-scale branches in an MSCB. In this way, the receptive-field mixture of the decoder becomes sample dependent rather than fixed across all images.

Directly replacing the original MSCB aggregation with a fully frequency-guided formulation could, however, unnecessarily perturb a stable multi-scale representation. To address this issue, we introduce a bounded residual guidance strategy. The frequency-guided branch aggregation is expressed as a controlled deviation from the conventional MSCB response,

$$M_{\text{RFG}} = M_{\text{ord}} + s (M_{\text{guide}} - M_{\text{ord}}),$$

where M_{ord} denotes the ordinary multi-scale aggregation, M_{guide} denotes the frequency-conditioned aggregation, and s is a bounded learnable guidance-strength parameter. This formulation preserves the conventional MSCB pathway as a reference and allows the model to progressively learn frequency-dependent deviations during optimization.

Based on this mechanism, we develop a ConvNeXt-Tiny U-shaped segmentation framework in which the deepest encoder feature is first enhanced using an existing FAFEM. The resulting low-/high-frequency descriptor is retained and transferred across network levels. After the first decoder feature is fused with the Stage-3 encoder representation, the proposed Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) uses this descriptor to generate sample-adaptive weights for the multi-kernel MSCB branches. The remaining decoder stages use lightweight conventional fusion and depthwise-separable convolution, keeping the additional architectural modification localized to the first encoder-decoder fusion stage.

The principal contributions of this work are summarized as follows:

- We introduce a frequency-conditioned multi-scale refinement mechanism, termed RFG-MSCB, which reuses the frequency descriptor generated by an existing bottleneck FAFEM to adaptively weight the multi-kernel branches of an existing MSCB. The contribution lies in the coupling between the frequency descriptor and the multi-scale branches rather than in FAFEM or MSCB themselves.
- We formulate the frequency-guided aggregation as a bounded residual deviation from the conventional MSCB response. This design preserves the ordinary multi-scale pathway as a reference while allowing a controlled, learnable, sample-dependent modification of branch importance.
- We introduce cross-level bottleneck-to-decoder frequency guidance by transferring the bottleneck frequency descriptor to the Stage-3 decoder refinement stage. This connects deep semantic frequency information with higher-resolution multi-scale feature processing while leaving the remaining decoder stages lightweight and unchanged.
- We evaluate the resulting ConvNeXt-Tiny segmentation architecture on multiple public colorectal polyp datasets and perform controlled ablation experiments to examine the contribution of frequency-aware enhancement, residual frequency guidance, and the selected Stage-3 placement.

Related work

Encoder–decoder architectures for medical image segmentation

Fully convolutional encoder–decoder networks remain a central design pattern for biomedical image segmentation. U-Net introduced a symmetric contracting and expanding architecture in which high-resolution encoder features are transferred to the decoder through skip connections, allowing semantic context and spatial detail to be combined during prediction [1]. UNet++ further showed that the structure of skip pathways is important by using nested dense connections to reduce the semantic discrepancy between encoder and decoder representations [2]. These works established the importance of hierarchical feature reuse, but they do not prescribe how the decoder should adapt its receptive-field mixture to the image-specific content of a polyp.

Polyp-specific convolutional models have explored efficient alternatives to large attention-based designs. HarDNet-MSEG combines a low-memory-traffic convolutional backbone with a lightweight decoder and demonstrates that strong polyp segmentation can be obtained without a transformer-heavy architecture [6]. PraNet introduced a parallel partial decoder and reverse attention to progressively refine uncertain polyp regions [7]. These methods highlight the value of efficient decoding and iterative refinement, but their main mechanisms are not designed to use frequency information from the bottleneck to regulate multi-scale convolutional branches at an earlier decoder stage.

Modern convolutional encoders provide another route to high-capacity dense prediction. ConvNeXt revisits convolutional network design with large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth, producing a hierarchical backbone with strong representation capacity [5]. In this work, ConvNeXt-Tiny is used as the encoder, while the novelty is concentrated in the decoder interaction between bottleneck frequency information and Stage-3 multi-scale convolution.

Transformer and contextual polyp segmentation

The large variation in polyp size, shape, color, and boundary contrast motivates architectures that capture both local appearance and broader anatomical context. Transformer-based approaches address this need by modeling long-range spatial relationships. TransUNet combines convolutional feature extraction with a transformer encoder for medical image segmentation [3], and Polyp-PVT applies pyramid vision transformer features and specialized fusion to polyp segmentation [4]. Recent polyp-segmentation studies have continued this direction through contrastive transformer modeling, contextual information flow, phase-wise feature pyramids, and pyramid-transformer feature fusion [8, 9, 10, 11].

These approaches show that global context and cross-level interaction are useful for polyp segmentation. However, attention-based or multi-stage contextual architectures can introduce additional architectural complexity. The present model therefore keeps a convolutional encoder–decoder form and asks a narrower question: whether a compact bottleneck frequency descriptor can guide the relative weighting of efficient multi-scale convolutional branches in the first decoder fusion stage.

Efficient multi-scale convolutional decoding

Multi-scale convolution is widely used because polyps may appear as small, subtle lesions or occupy a large image region. Parallel kernels and dilated or depthwise branches allow a decoder to aggregate local and broader responses without relying entirely on self-attention. EMCAD introduced an efficient multi-scale convolutional attention decoder for medical image segmentation and uses Multi-Scale Convolution Blocks (MSCBs) as part of its lightweight decoding design [12]. In an MSCB, multiple convolutional branches provide different receptive fields and are combined to form a multi-scale response.

MSCB is prior work and is not introduced in this paper. The limitation relevant to the present method is that a conventional MSCB aggregation treats the branch mixture as fixed by the block design rather than conditioning the branch weights on the image-specific frequency characteristics extracted at the bottleneck. Our RFG-MSCB retains the ordinary MSCB path as a reference and adds only a bounded, frequency-conditioned residual deviation from that path.

Frequency-aware and adaptive feature recalibration

Attention and recalibration mechanisms provide a general precedent for using compact descriptors to control feature responses. Squeeze-and-Excitation Networks use global channel descriptors to recalibrate convolutional features [13], while Selective Kernel Networks adaptively weight convolutional branches with different receptive fields [14]. These methods are important novelty comparators because they show that descriptor-driven reweighting and adaptive receptive-field selection are established ideas.

Frequency-aware attention provides a closer conceptual link. FcaNet interprets channel attention through frequency analysis and uses frequency components for channel attention [15]. In polyp segmentation, Polyp-Mamba combines multi-frequency perception with a gated selection mechanism to model frequency-sensitive representations [16]. These studies motivate the usefulness of frequency information, but they do not correspond to the specific coupling used here: a frequency descriptor produced by a bottleneck FAFEM is transferred across levels and used to regulate Stage-3 MSCB branch weights.

The Frequency-Aware Feature Enhancement Module (FAFEM) is also treated as prior work in this paper. It is adopted at the bottleneck to enhance deep features and produce a low-/high-frequency descriptor. The contribution of this work is not FAFEM itself, but the reuse of its descriptor as cross-level guidance for multi-scale convolution in the decoder. The exact original FAFEM citation should be inserted once the source title, DOI, arXiv identifier, or publisher record is verified.

Boundary and recent polyp-segmentation refinements

Weak and visually ambiguous lesion boundaries remain a major challenge in polyp segmentation. Boundary-aware methods often introduce explicit edge supervision, auxiliary contour branches, image-pyramid high-frequency cues, or reverse attention. MEGANet employs multi-scale edge-guided attention for weak-boundary polyp segmentation [17], and MSBP-Net performs multi-scale boundary prediction to improve contour delineation [18]. Other recent methods model uncertainty, contextual flow, frequency-sensitive representations, and multi-stage refinement [19, 9, 16, 20].

The proposed architecture does not add an explicit boundary branch, edge label, or boundary loss. Boundary accuracy is instead expected to benefit indirectly from adaptive multi-scale decoding and from the use of frequency-guided branch weights at the first decoder fusion stage. Consequently, claims about boundary improvement should be supported by the experimental results rather than by describing a dedicated boundary module.

Positioning of the proposed method

Existing work has improved polyp segmentation through U-shaped skip reuse, reverse attention, transformer-based context modeling, efficient convolutional decoding, frequency-aware representation learning, and boundary-aware refinement. The proposed method is positioned at the intersection of efficient multi-scale convolution and frequency-aware recalibration.

Specifically, the final model uses a ConvNeXt-Tiny encoder, a bottleneck FAFEM, a lightweight residual bottleneck, Stage-3 SimpleFusion, the proposed Residual Frequency-Guided MSCB, and standard remaining decoder stages. FAFEM and MSCB are reused prior modules. The proposed contribution is the coupling between them: the bottleneck FAFEM frequency descriptor adaptively weights MSCB branches at Stage 3, the guided response is introduced as a bounded residual deviation from ordinary MSCB aggregation, and the guidance forms a bottleneck-to-Stage-3 cross-level connection. This framing separates the new RFG-MSCB mechanism from the prior components on which it is built.

Method

Overall Architecture

The proposed framework follows a U-shaped encoder-decoder architecture built upon a ConvNeXt-Tiny encoder. The central objective is to exploit frequency information extracted from deep semantic features to adaptively regulate multi-scale feature processing during decoding. Rather than independently applying frequency enhancement and multi-scale convolution, the proposed model establishes a cross-level interaction between the bottleneck and the first decoder stage.

Let

$$I \in \mathbb{R}^{3 \times H \times W}$$

denote an input colonoscopy image. The ConvNeXt-Tiny encoder extracts four hierarchical feature maps,

$$\{F_1, F_2, F_3, F_4\} = E(I),$$

where

$$F_1 \in \mathbb{R}^{96 \times H/4 \times W/4}, \quad F_2 \in \mathbb{R}^{192 \times H/8 \times W/8},$$

$$F_3 \in \mathbb{R}^{384 \times H/16 \times W/16}, \quad F_4 \in \mathbb{R}^{768 \times H/32 \times W/32}.$$

For an input resolution of 352×352 , their spatial dimensions are 88×88 , 44×44 , 22×22 , and 11×11 , respectively.

The deepest encoder feature F_4 is first processed by a Frequency-Aware Feature Enhancement Module (FAFEM), which produces an enhanced bottleneck representation together with a compact frequency descriptor. The enhanced representation is passed through a lightweight residual bottleneck, whereas the frequency descriptor is propagated to the first decoder stage. After the decoder feature is fused with F_3 , a Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) uses the frequency descriptor to adaptively regulate its multi-scale convolution branches.

The overall processing can therefore be summarized as

$$(\tilde{F}_4, \mathbf{z}_f) = \mathcal{F}_{\text{FAFEM}}(F_4),$$

$$B = \mathcal{B}(\widetilde{F}_4),$$

$$D_4 = \mathcal{M}_{\text{RFG}}(\mathcal{S}_3[\mathcal{D}_4(B), F_3], \mathbf{z}_f),$$

followed by

$$D_3 = \mathcal{S}_2[\mathcal{D}_3(D_4), F_2],$$

$$D_2 = \mathcal{S}_1[\mathcal{D}_2(D_3), F_1],$$

$$D_1 = \mathcal{D}_1(D_2).$$

Here, $\mathcal{D}_i$ denotes a decoder block, $\mathcal{S}_i$ denotes the corresponding skip-feature fusion operation, $\mathbf{z} * f$ is the frequency descriptor extracted at the bottleneck, and $\mathcal{M} * \text{RFG}$ denotes the proposed residual frequency-guided multi-scale refinement mechanism.

The final prediction head transforms D_1 into a single-channel segmentation logit map, which is bilinearly resized to the original image resolution.

ConvNeXt-Tiny Encoder

The encoder employs ConvNeXt-Tiny with stage depths

$$[3, 3, 9, 3]$$

and channel dimensions

$$[96, 192, 384, 768].$$

Before entering the encoder, each RGB image is normalized using the ImageNet statistics

$$\mu = (0.485, 0.456, 0.406), \quad \sigma = (0.229, 0.224, 0.225).$$

The first stage uses a 4×4 convolution with stride 4 to generate F_1 . Each subsequent stage begins with spatial downsampling using a 2×2 stride-2 convolution.

For an input feature X , a ConvNeXt block first performs a 7×7 depthwise convolution. The resulting feature is normalized in channels-last format and processed by a pointwise multilayer perceptron with a fourfold channel expansion:

$$Z = \text{DWConv}_{7 \times 7}(X),$$

$$V = W_2(\text{GELU}(W_1(\text{LN}(Z)))).$$

The block output is obtained through layer-scaled residual learning,

$$Y = X + \text{DropPath}(\gamma \odot V),$$

where γ is a learnable layer-scale parameter initialized to 10^{-6} .

A stochastic-depth rate of 0.25 is used throughout the encoder. Feature dropout with probability 0.2 is additionally applied after the second, third, and fourth encoder stages.

Frequency-Aware Bottleneck Enhancement

The deepest encoder representation F_4 contains strong semantic information but has undergone substantial spatial downsampling. To expose complementary frequency characteristics of this representation, FAFEM is applied exclusively at the bottleneck.

Given

$$X = F_4,$$

a local low-frequency approximation is obtained using 3×3 average pooling,

$$X_L = \text{AvgPool}_{3 \times 3}(X),$$

and the corresponding high-frequency component is defined as

$$X_H = X - X_L.$$

The low- and high-frequency components are independently transformed using depthwise convolutional branches,

$$\widehat{X}_L = \phi_L(X_L), \quad \widehat{X}_H = \phi_H(X_H),$$

where each $\phi(\cdot)$ contains a 3×3 depthwise convolution followed by GELU activation.

The two components are concatenated and globally pooled to obtain the frequency descriptor

$$\mathbf{z}_f = \text{GAP}([\widehat{X}_L, \widehat{X}_H]) \in \mathbb{R}^{2C}.$$

Since $C = 768$ at the bottleneck,

$$\mathbf{z}_f \in \mathbb{R}^{1536}.$$

A lightweight channel-gating network generates adaptive weights for the low- and high-frequency representations. Their weighted combination is spatially refined and introduced through a residual connection:

$$\widetilde{F}_4 = F_4 + \gamma_f \mathcal{R}_f(w_L \odot \widehat{X}_L + w_H \odot \widehat{X}_H),$$

where γ_f is a learnable scalar initialized to zero.

Importantly, FAFEM itself is not introduced in this work. In the proposed architecture, its frequency descriptor $\mathbf{z}_f$ is additionally reused to condition multi-scale processing at a later decoder stage.

Lightweight Residual Bottleneck

The frequency-enhanced representation $\tilde{F}_4$ is subsequently processed by a lightweight residual bottleneck.

A 1×1 convolution first reduces the number of channels from 768 to 192:

$$X_r = \text{GELU} \left(\text{LN} \left(\text{Conv}_{1 \times 1}(\tilde{F}_4) \right) \right).$$

A 3×3 depthwise-separable convolution performs spatial transformation in the reduced feature space. The representation is subsequently projected back to 768 channels.

The bottleneck output is formulated as

$$B = \tilde{F}_4 + \gamma_b \mathcal{H}(\tilde{F}_4),$$

where $\mathcal{H}(\cdot)$ denotes the bottleneck transformation and γ_b is a learnable scalar initialized to zero.

This initialization makes the bottleneck behave initially as an identity mapping and allows its transformation to be progressively introduced during optimization.

Residual Frequency-Guided Multi-Scale Convolution

The principal architectural contribution of this work is the Residual Frequency-Guided Multi-Scale Convolution mechanism.

After the bottleneck feature is upsampled from $H/32$ to $H/16$, the decoder representation is reduced from 768 to 384 channels and concatenated with the Stage-3 encoder feature F_3 . A lightweight fusion block transforms the resulting 768-channel representation into

$$X_3 \in \mathbb{R}^{384 \times H/16 \times W/16}.$$

For a 352×352 input,

$$X_3 \in \mathbb{R}^{384 \times 22 \times 22}.$$

A conventional lightweight multi-scale convolution block first expands the feature dimension from C to $2C$:

$$X_e = \text{ReLU6}(\text{BN}(\text{Conv}_{1 \times 1}(X_3))).$$

Three parallel depthwise convolution branches with kernel sizes

$$k \in \{1, 3, 5\}$$

then generate multi-scale responses,

$$B_k = \phi_k(X_e),$$

where each ϕ_k consists of a depthwise convolution, Batch Normalization, and ReLU6 activation.

In a conventional MSCB, these branches are combined uniformly:

$$M_{\text{ord}} = B_1 + B_3 + B_5.$$

Instead of treating the three receptive fields identically for every image, the proposed mechanism conditions their relative contributions on the bottleneck frequency descriptor $\mathbf{z}_f$.

A lightweight guidance network maps the 1536-dimensional descriptor to three branch logits:

$$\mathbf{q} = W_{g2}(\text{GELU}(W_{g1}\mathbf{z}_f)),$$

where

$$W_{g1} : 1536 \rightarrow 48, \quad W_{g2} : 48 \rightarrow 3.$$

The normalized branch weights are calculated as

$$[\alpha_1, \alpha_3, \alpha_5] = 3 \text{Softmax}(\mathbf{q}),$$

such that

$$\alpha_1 + \alpha_3 + \alpha_5 = 3.$$

The frequency-guided response becomes

$$M_{\text{guide}} = \alpha_1 B_1 + \alpha_3 B_3 + \alpha_5 B_5.$$

Directly replacing the conventional multi-scale response with the guided response may excessively perturb an already effective MSCB representation. We therefore introduce the guidance as a bounded residual deviation:

$$M_{\text{RFG}} = M_{\text{ord}} + s(M_{\text{guide}} - M_{\text{ord}}),$$

where

$$0 \leq s \leq 0.5.$$

The learnable guidance-strength parameter s is initialized to 0.05. Consequently, the frequency-conditioned branch regulation is introduced conservatively during early optimization.

Furthermore, the final guidance-layer weights and biases are initialized such that the initial branch logits are equal. Therefore,

$$\alpha_1 = \alpha_3 = \alpha_5 = 1$$

at initialization, which gives

$$M_{\text{guide}} = M_{\text{ord}}.$$

The module thus initially reproduces the conventional multi-scale path, while optimization is allowed to learn image-dependent deviations from uniform branch weighting.

The resulting feature is processed using channel shuffle and a 1×1 projection from $2C$ back to C , followed by Batch Normalization. A final residual connection preserves the input representation:

$$Y_3 = X_3 + \mathcal{P}(\text{Shuffle}(M_{\text{RFG}})).$$

This formulation establishes an explicit cross-level connection between deep frequency information extracted at the bottleneck and multi-scale spatial refinement performed at $H/16$ resolution.

Progressive Decoder and Skip Fusion

After frequency-guided Stage-3 refinement, the decoder progressively restores spatial resolution using bilinear upsampling and depthwise-separable convolution.

The first refined decoder feature has 384 channels:

$$D_4 = Y_3.$$

The subsequent decoder stages are

$$D_3 = \mathcal{S}_2[\mathcal{D}_3(D_4), F_2],$$

$$D_2 = \mathcal{S}_1[\mathcal{D}_2(D_3), F_1],$$

followed by

$$D_1 = \mathcal{D}_1(D_2).$$

Their dimensions for a 352×352 input are

$$D_4 : 384 \times 22 \times 22,$$

$$D_3 : 192 \times 44 \times 44,$$

$$D_2 : 96 \times 88 \times 88,$$

and

$$D_1 : 96 \times 176 \times 176.$$

At the F_2 and F_1 skip levels, the upsampled decoder representation is concatenated with the corresponding encoder feature and processed by a lightweight fusion block containing a 1×1 projection followed by a depthwise-separable 3×3 convolution.

No additional attention mechanism, frequency module, MSCB, or boundary-specific refinement block is applied at these decoder stages.

The final decoder stage has no direct encoder skip connection because the ConvNeXt stem begins at one-quarter of the input resolution.

Prediction Head

The final decoder feature D_1 is processed using a lightweight refinement head. Two depthwise-separable 3×3 convolutions transform the feature dimensions as

$$96 \rightarrow 96 \rightarrow 48.$$

Each transformation is followed by Layer Normalization and GELU activation. A final 1×1 convolution produces a single-channel segmentation logit map:

$$P_0 = \text{Conv}_{1 \times 1}(\mathcal{R}(D_1)).$$

The prediction is then bilinearly resized from $H/2 \times W/2$ to the original input resolution:

$$P = \text{Up}_{H,W}(P_0).$$

The network directly returns logits. During inference, the foreground probability map is obtained as

$$\hat{Y} = \sigma(P),$$

where $\sigma(\cdot)$ denotes the sigmoid function.

The selected architecture does not employ auxiliary prediction heads or deep supervision.

Model Complexity

The complete proposed model contains

$$30,057,415$$

trainable parameters, corresponding to approximately

$$30.06 \text{ M}$$

parameters.

The additional frequency-guided mechanism is introduced only at the first decoder–encoder fusion stage. The remaining decoder retains lightweight depthwise-separable convolution and conventional skip fusion, limiting the architectural modification to the stage at which high-level semantic information is first combined with the relatively high-resolution F_3 representation.

Experiments

Datasets and Data Splits

We used Kvasir-SEG [21], CVC-ClinicDB [22], CVC-300 [25], CVC-ColonDB [24], and ETIS-LaribPolypDB [23]. Kvasir-SEG and CVC-ClinicDB were split independently with a 90:10 ratio using shuffled sampling with random seed 42. Their training subsets were merged, giving 900 Kvasir-SEG and 550 CVC-ClinicDB images for training. The corresponding held-out subsets of 100 and 62 images were merged for validation and were also evaluated separately for reporting. CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB were used only for external evaluation; none influenced training or checkpoint selection.

Table 4: Dataset use in the final experiment.

Dataset	Total	Train	Validation	External test
Kvasir-SEG	1000	900	100	–
CVC-ClinicDB	612	550	62	–
CVC-300	60	–	–	60
CVC-ColonDB	380	–	–	380
ETIS-LaribPolypDB	196	–	–	196

Images were converted to RGB, resized to 352×352 with bilinear interpolation, and scaled to $[0, 1]$. Masks were resized with nearest-neighbor interpolation and binarized. The encoder applied ImageNet normalization with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

Implementation Details

The final seed-42 experiment was implemented in PyTorch 2.5.1+cu124 and trained for 200 epochs with batch size 24. AdamW with betas (0.9, 0.999) used separate parameter groups. The maximum learning rates were 3.0×10^{-4} for the decoder and 4.5×10^{-4} for the final refinement head. Encoder Stages 4, 3, 2, and 1 plus the stem used 3.0×10^{-5} , 2.4×10^{-5} , 1.92×10^{-5} , and 1.536×10^{-5} , respectively, corresponding to layer-wise decay 0.8. Weight decay was 1.0×10^{-4} for the decoder, 5.0×10^{-2} for the encoder, and 1.0×10^{-2} for the refinement head.

Learning rates were warmed up linearly for five epochs from 0.1 to 1.0 of each group maximum and then cosine-annealed over the remaining epochs to 1.0×10^{-6} . Training used AMP FP16, global gradient clipping at norm 1.0, and an exponential moving average (EMA) with decay 0.995. All parameters were trainable from the first epoch; no staged freezing or decoder-only warmup was used. The encoder was initialized from `convnext_tiny_22k_1k_384.pth`, logged as ImageNet pretrained weights. Deep supervision and frequency augmentation were disabled, and sampling was uniform. Early stopping was configured with patience 30 from the start of training but did not trigger; training completed at the maximum epoch. The final model has 30,057,415 trainable parameters.

Training augmentation applied horizontal and vertical flips independently with probability 0.5, rotation in $[-15^\circ, 15^\circ]$ with probability 0.5, and a random crop ratio in $[0.7, 1.0]$ followed by resizing with probability 0.5. Joint brightness and contrast factors in $[0.85, 1.15]$ and saturation in $[0.9, 1.1]$ were applied with probability 0.5. A 3×3 Gaussian blur and additive Gaussian noise with standard deviation 0.015 were applied with probabilities 0.15 and 0.25, respectively.

Loss Function

The final objective was

$$\mathcal{L} = 0.55\mathcal{L}_{\text{Dice}} + 0.25\mathcal{L}_{\text{BCE}} + 0.20\mathcal{L}_{\text{boundary}}. \quad (1)$$

The soft Dice term used a smoothing constant of 1.0. The BCE-with-logits term used targets smoothed as $y_{\text{BCE}} = 0.98y + 0.01$, corresponding to label smoothing 0.02. The boundary-aware term used $\kappa = 5$. The available focal-Tversky term had weight zero and therefore did not contribute to training.

Evaluation Metrics

We report mean Dice (mDice), mean intersection over union (mIoU), S-measure S_α [26], weighted F-measure F_β^w , mean and maximum E-measure (mE_ϕ and $\max E_\phi$) [27], and mean absolute error (MAE) [28]. For each image, the thresholded prediction P and binary ground truth G were flattened and evaluated as

$$\text{Dice}(P, G) = \frac{2|P \cap G| + 10^{-6}}{|P| + |G| + 10^{-6}}, \quad \text{IoU}(P, G) = \frac{|P \cap G| + 10^{-6}}{|P| + |G| - |P \cap G| + 10^{-6}}. \quad (2)$$

The reported mDice and mIoU are arithmetic means of these per-image scores, rather than global pixel aggregates. S-measure, weighted F-measure, the E-measure curve, and MAE were computed per image with `py_sod_metrics` from 8-bit probability maps and binary ground truth. The mean and maximum of the resulting E-measure curve were reported.

Model Selection and Evaluation Protocol

At each epoch, EMA weights were evaluated on the merged held-out Kvasir-SEG and CVC-ClinicDB validation set. Probabilities were obtained with one sigmoid operation and binarized using the strict rule $p > 0.45$. A checkpoint was saved only when validation IoU was strictly greater than the previous best. The selected checkpoint was epoch 184, with merged-validation IoU 0.8780511022 and Dice 0.9348011783. Final evaluation loaded this best checkpoint strictly and evaluated the two held-out subsets separately, followed by the full CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB sets. Test-time augmentation was disabled, and external test results did not affect checkpoint selection.

Comparative study

Quantitative analysis

Qualitative analysis

Models	Seen		ClinicDB		Unseen		ColonDB		ETIS		Epoch
	Kvasir-Seg				CVC-300						
	Dice	iou	Dice	iou	Dice	iou	Dice	iou	Dice	iou	
HCA-Net	93.6	89.7	95.6	90.9	91.2	87.3	83.4	76.9	83.3	78.9	
Our9388	93.49	87.77	94.92	90.33	90.33	82.52	79.74	69.20	88.69	79.67	100
Ours	92.58	86.41	95.31	91.04	91.49	84.32	83.34	73.75	87.39	77.61	1370
Ours-93.04	92.30				91.25	83.91			84.96	73.85	
Ours-93.66	92.75	86.48	95.20	90.83	91.61	84.52	82.81	72.96	85.39	74.50	2836

Table 1 Quantitative comparison of various models on the Kvasir-SEG dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	F ^w	S	mE	maxE	MAE
CTNet [8]	2024	0.917	0.863	0.910	0.928	0.959	0.964	0.023
MEGANet [17]	2024	0.913	0.863	0.907	0.918	0.956	0.959	0.025
CAFÉ-Net [29]	2024	0.933	0.889	0.927	0.939	0.967	?	0.019
Polyp-LVT [30]	2024	0.909	0.851	0.913	0.922	0.941	?	0.024
Polyp-Mamba [16]	2025	0.919	0.867	0.912	0.951	-	0.968	0.021
MEIN [19]	2025	0.905	0.847	0.905	0.914	-	0.964	0.026
MF-Net [11]	2025	0.915	0.863	0.939	0.924	0.962	?	0.022
MSBP-Net [18]	2025	0.919	0.868	0.916	0.926	0.961	?	0.023
CIFFormer [9]	2025	0.926	0.876	0.929	0.939	0.967	?	0.019
PFPRNet [10]	2025	0.930	0.881	0.920	0.935	0.971	?	0.019
PraNet-V2 [20]	2025	0.907	0.853	?	?	?	?	0.025
MLB-Net [31]	2026	0.926	0.878	0.921	0.938	0.969	0.975	0.021
MSPN [32]	2026	0.927	0.879	?	?	?	?	0.021

Methods	Year	mDice	mIOU	F^w	S	mE	maxE	MAE
DVFIP-Net [33]	2026	0.922	0.874	?	?	?	?	?
Ours	-	0.935	0.879	0.901	0.935	0.952	0.963	0.025

Table 2 Quantitative comparison of various models on the CVC-ClinicDB dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	F^w	S	mE	maxE	MAE
CTNet [8]	2024	0.936	0.887	0.934	0.952	0.983	0.986	0.006
MEGANet [17]	2024	0.938	0.894	0.940	0.950	-	0.987	0.007
Polyp-Mamba [16]	2025	0.941	0.896	0.936	0.970	-	0.970	0.008
MEIN [19]	2025	0.923	0.878	0.928	0.944	-	0.970	0.008
MF-Net [11]	2025	0.931	0.884	0.953	0.949	0.983	?	0.006
MSBP-Net [18]	2025	0.940	0.892	0.927	0.939	0.976	?	0.007
CIFFormer [9]	2025	0.944	0.897	0.941	0.934	0.972	?	0.007
PFPRNet [10]	2025	0.949	0.903	0.947	0.957	0.989	?	0.006
PraNet-V2 [20]	2025	0.929	0.88	?	?	?	?	0.009
MLB-Net [31]	2026	0.945	0.902	0.948	0.958	0.989	0.992	0.006
MSPN [32]	2026	0.943	0.902	?	?	?	?	0.006
DVFIP-Net [33]	2026	0.942	0.896	?	?	?	?	?
Ours	-	0.951	0.906	0.931	0.954	0.983	0.990	0.009

Table 3 Quantitative comparison of various models on the ETIS-LaribPolypDB dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	F^w	S	mE	maxE	MAE
CTNet [8]	2024	0.810	0.734	0.776	0.886	0.913	0.921	0.014
MEGANet [17]	2024	0.789	0.709	0.753	0.866	0.855	0.915	0.015
Polyp-Mamba [16]	2025	0.756	0.668	0.756	0.887	-	0.872	0.014
MEIN [19]	2025	0.788	0.711	0.764	0.870	-	0.902	0.018
MF-Net [11]	2025	0.821	0.750	0.850	0.901	0.925	?	0.011
MSBP-Net [18]	2025	0.795	0.718	0.770	0.873	0.899	?	0.019
CIFFormer [9]	2025	0.810	0.746	0.858	0.889	0.918	?	0.013
PFPRNet [10]	2025	0.828	0.764	0.839	0.908	0.935	?	0.012
PraNet-V2 [20]	2025	0.713	0.633	?	?	?	?	0.021
MLB-Net [31]	2026	0.821	0.744	0.780	0.899	0.914	0.935	0.013
MSPN [32]	2026	0.798	0.717	?	?	?	?	0.017
DVFIP-Net [33]	2026	0.811	0.730	?	?	?	?	?
Ours	-	0.887	0.797	0.730	0.873	0.883	0.913	0.014

Table 4 Quantitative comparison of various models on the CVC-ColonDB dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	F^w	S	mE	maxE	MAE
CTNet [8]	2024	0.813	0.734	0.801	0.874	0.915	0.919	0.040
MEGANet [17]	2024	0.793	0.714	0.779	0.854	0.892	0.895	0.034
Polyp-Mamba [16]	2025	0.791	0.713	0.765	0.846	-	0.902	0.034
MEIN [19]	2025	0.791	0.715	0.785	0.856	-	0.897	0.034
MF-Net [11]	2025	0.825	0.751	0.864	0.887	0.929	?	0.024
MSBP-Net [18]	2025	0.810	0.731	0.810	0.861	0.912	?	0.029
CIFFormer [9]	2025	0.823	0.741	0.851	0.874	0.918	?	0.026
PFPRNet [10]	2025	0.819	0.741	0.843	0.882	0.915	?	0.028
PraNet-V2 [20]	2025	0.702	0.620	?	?	?	?	0.044
MLB-Net [31]	2026	0.832	0.757	0.825	0.886	0.924	0.931	0.023
MSPN [32]	2026	0.811	0.734	?	?	?	?	0.029
DVFIP-Net [33]	2026	0.826	0.746	?	?	?	?	?
Ours	-	79.74	69.20	0.759	0.869	0.885	0.903	0.035

Table 5 Quantitative comparison of various models on the CVC-300(Endoscene) dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	F^w	S	mE	maxE	MAE
CTNet [8]	2024	0.908	0.844	0.894	0.975	0.975	0.982	0.006
MEGANet [17]	2024	0.899	0.834	0.882	0.935	0.966	0.969	0.007
Polyp-Mamba [16]	2025	0.906	0.840	0.888	0.936	-	0.975	0.006
MEIN [19]	2025	0.893	0.828	0.880	0.930	-	0.965	0.007
MF-Net [11]	2025	0.897	0.836	0.918	0.941	0.966	?	0.007
MSBP-Net [18]	2025	0.903	0.859	0.927	0.942	0.967	?	0.019
CIFFormer [9]	2025	0.883	0.822	0.912	0.939	0.961	?	0.10
PFPRNet [10]	2025	0.891	0.833	0.917	0.928	0.950	?	0.013
PraNet-V2 [20]	2025	0.873	0.795					0.007
MLB-Net [31]	2026	0.915	0.852	0.909	0.974	0.974	0.978	0.005
MSPN [32]	2026	0.907	0.840					0.006
DVFIP-Net [33]	2026	0.900	0.833	?	?	?	?	?
Ours	-	0.916	0.845	0.848	0.939	0.951	0.969	0.007

Discussion

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

References

- [1] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*,

pages 234–241. Springer, 2015.

- [2] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. UNet++: Redesigning skip connections to exploit multiscale features in image segmentation. *IEEE Transactions on Medical Imaging*, 39(6):1856–1867, 2020.
- [3] Jieneng Chen, Yongyi Lu, Qihang Yu, Xiangde Luo, Ehsan Adeli, Yan Wang, Le Lu, Alan L. Yuille, and Yuyin Zhou. TransUNet: Transformers make strong encoders for medical image segmentation, 2021.
- [4] Bo Dong, Wenhai Wang, Deng-Ping Fan, Jinpeng Li, Huazhu Fu, and Ling Shao. Polyp-PVT: Polyp segmentation with pyramid vision transformers. *CAAI Artificial Intelligence Research*, page 9150015, 2023.
- [5] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11966–11976, 2022.
- [6] Chien-Hsiang Huang, Hung-Yu Wu, and Youn-Long Lin. HarDNet-MSEG: A simple encoder-decoder polyp segmentation neural network that achieves over 0.9 mean dice and 86 fps, 2021.
- [7] Deng-Ping Fan, Ge-Peng Ji, Tao Zhou, Geng Chen, Huazhu Fu, Jianbing Shen, and Ling Shao. PraNet: Parallel reverse attention network for polyp segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2020*, pages 263–273. Springer, 2020.
- [8] Bin Xiao, Jinwu Hu, Weisheng Li, Chi-Man Pun, and Xiuli Bi. CTNet: Contrastive transformer network for polyp segmentation. *IEEE Transactions on Cybernetics*, 54:5040–5053, 2024.
- [9] Cunlu Xu, Long Lin, Bin Wang, and Jun Liu. CIFORMER: A contextual information flow guided transformer for colorectal polyp segmentation. *Neurocomputing*, 644:130413, 2025.
- [10] Jinghui Chu, Wangtao Liu, Qi Tian, and Wei Lu. PFPRNet: A phase-wise feature pyramid with retention network for polyp segmentation. *IEEE Journal of Biomedical and Health Informatics*, 29:1137–1150, 2025.
- [11] Shuangyuan Wang, Shengmao Lin, Fujia Sun, and Xiaobo Li. Multi-feature fusion for accurate polyp segmentation using pyramid visual transformers. *Expert Systems with Applications*, 280:127558, 2025.
- [12] Md Mostafijur Rahman, Mustafa Munir, and Radu Marculescu. EMCAD: Efficient multi-scale convolutional attention decoding for medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11769–11779, 2024.
- [13] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7132–7141, 2018.
- [14] Xiang Li, Wenhai Wang, Xiaolin Hu, and Jian Yang. Selective kernel networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 510–519, 2019.
- [15] Zequn Qin, Pengyi Zhang, Fei Wu, and Xi Li. FcaNet: Frequency channel attention networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 763–772, 2021.
- [16] Xingguo Zhu, Wei Wang, Chen Zhang, and Haifeng Wang. Polyp-Mamba: A hybrid multi-frequency perception gated selection network for polyp segmentation. *Information Fusion*, 115:102759, 2025.
- [17] Nhat-Tan Bui, Dinh-Hieu Hoang, Quang-Thuc Nguyen, Minh-Triet Tran, and Ngan Le. MEGANet: Multi-scale edge-guided attention network for weak boundary polyp segmentation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 7985–7994, 2024.
- [18] Xing-Liang Pan, Ju-Rong Ding, Xia Li, Shuo Liu, Jie Wang, Bo Hua, Guo-Zhi Tang, and Chang-Hua Zhong. MSBP-Net: A multi-scale boundary prediction network for automated polyp segmentation. *Pattern Recognition*, 170:112101, 2026.
- [19] Xiaolu Kang, Zhuoqi Ma, Kang Liu, Yunan Li, and Qiguang Miao. Modeling multi-scale uncertainty with evidence integration for reliable polyp segmentation. *Neural Networks*, 189:107553, 2025.

- [20] Bo-Cheng Hu, Ge-Peng Ji, Dian Shao, and Deng-Ping Fan. PraNet-V2: Dual-supervised reverse attention for medical image segmentation, 2025.
- [21] Debesh Jha, Pia H. Smedsrud, Michael A. Riegler, Pål Halvorsen, Thomas de Lange, Dag Johansen, and Håvard D. Johansen. Kvasir-SEG: A segmented polyp dataset. In *MultiMedia Modeling*, pages 451–462. Springer, 2020.
- [22] Jorge Bernal, F. Javier Sánchez, Gloria Fernández-Esparrach, Debora Gil, Cristina Rodríguez, and Fernando Vilariño. WM-DOVA maps for accurate polyp highlighting in colonoscopy: Validation versus saliency maps from physicians. *Computerized Medical Imaging and Graphics*, 43:99–111, 2015.
- [23] Juan Silva, Aymeric Histace, Olivier Romain, Xavier Dray, and Bertrand Granado. Toward embedded detection of polyps in WCE images for early diagnosis of colorectal cancer. *International Journal of Computer Assisted Radiology and Surgery*, 9(2):283–293, 2014.
- [24] Nima Tajbakhsh, Suryakanth R. Gurudu, and Jianming Liang. Automated polyp detection in colonoscopy videos using shape and context information. *IEEE Transactions on Medical Imaging*, 35(2):630–644, 2016.
- [25] David Vázquez, Jorge Bernal, F. Javier Sánchez, Gloria Fernández-Esparrach, Antonio M. López, Adriana Romero, Michal Drozdal, and Aaron Courville. A benchmark for endoluminal scene segmentation of colonoscopy images. *Journal of Healthcare Engineering*, 2017:1–9, 2017.
- [26] Deng-Ping Fan, Ming-Ming Cheng, Yun Liu, Tao Li, and Ali Borji. Structure-measure: A new way to evaluate foreground maps. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 4558–4567, 2017.
- [27] Deng-Ping Fan, Cheng Gong, Yang Cao, Bo Ren, Ming-Ming Cheng, and Ali Borji. Enhanced-alignment measure for binary foreground map evaluation. In *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence*, pages 698–704, 2018.
- [28] Federico Perazzi, Philipp Krähenbühl, Yael Pritch, and Alexander Hornung. Saliency filters: Contrast based filtering for salient region detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 733–740, 2012.
- [29] CAFÉ-Net, 2024. Incomplete manuscript citation; authors and venue require manual verification.
- [30] Polyp-LVT, 2024. Incomplete manuscript citation; authors and venue require manual verification.
- [31] MLB-Net, 2026. Incomplete manuscript citation; authors and venue require manual verification.
- [32] From coarse to fine: Multi-stage progressive network for colon polyp segmentation, 2026. Incomplete manuscript citation; authors and venue require manual verification.
- [33] DVFIP-Net, 2026. Incomplete manuscript citation; authors and venue require manual verification.
- [34] HCA-Net. Incomplete manuscript reference placeholder; authors, venue, and year require manual verification.

Methods	Year	Param (M)	GFLOPs (G)
CTNet [8]	2024		
MEGANet [17]	2024		
Polyp-Mamba [16]	2025	49.5	27.9
MEIN [19]	2025		
MF-Net [11]	2025		
MSBP-Net [18]	2025	25.52	12.86
CIFFormer [9]	2025	32.44	13.14
PFPRNet [10]	2025	118.52	39.39
PraNet-V2 [20]	2025		
MLB-Net [31]	2026	87.70	

Methods	Year	Param (M)	GFLOPs (G)
MSPN [32]	2026	28.56	13.21
DVFIP-Net [33]	2026	46.62	19.37
Ours	-	29.60	14.57

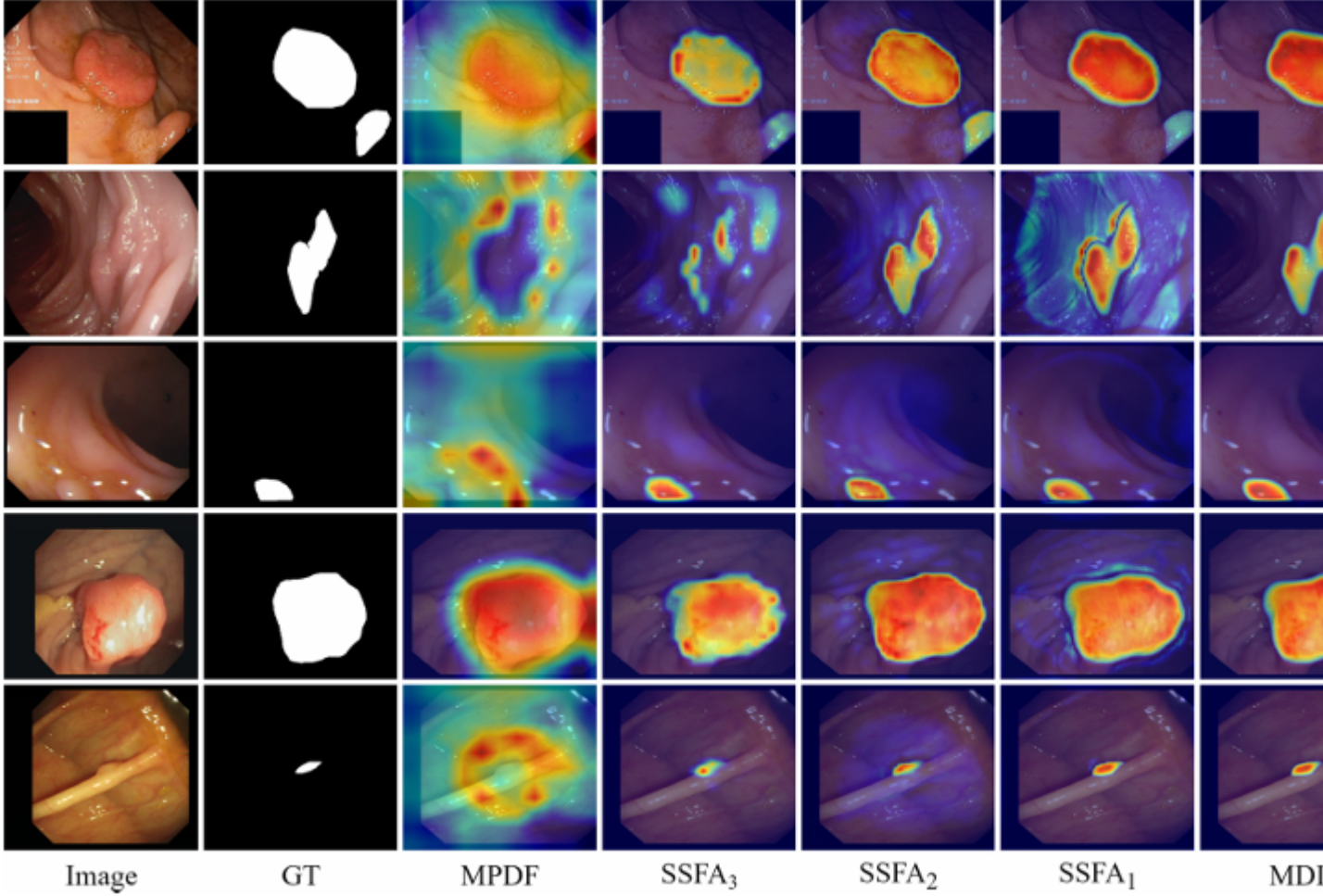


Fig. 6 Attention heatmaps for the model's MPDF, SSFAi, and MDI modules. Darker colors indicate higher attention

- **MultiScaleContext**: tests whether multi-scale/global context helps.
- **BSEI**: tests boundary-aware feature enhancement.
- **DetailBranch**: tests high-resolution spatial detail recovery.
- **GatedDetailFusion**: tests whether adaptive fusion is better than simple fusion.



Fig. 1 Six examples of polyp images, with the circled green area in each image representing the polyp

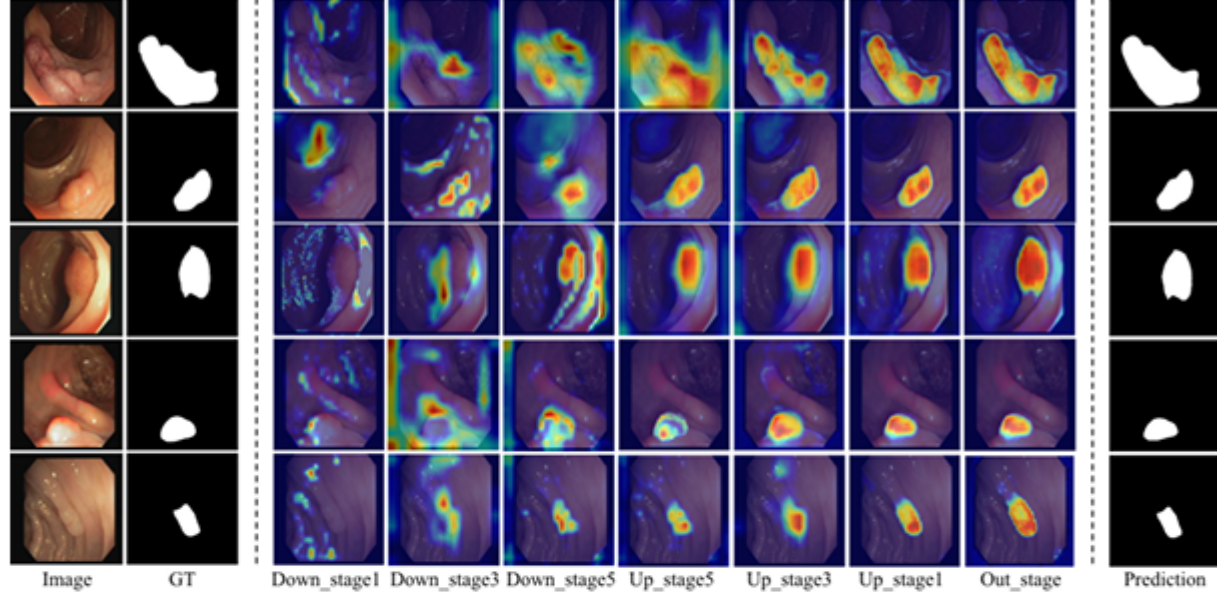


Fig. 8. Module visualization feature map.

TABLE VII
ABLATION STUDY ON BACKBONE WIDTH (BASE_CHANNEL) AND THE COMPUTATIONAL OVERHEAD OF HCA.

base_channel	HCA	CVC	ClinicDB	ETIS-LaribPolypDB		Param (MB)	GFLOPs (G)
		mDice	mIoU	mDice	mIoU		
7	-	0.801	0.748	0.662	0.627	3.45	2.15
7	+	0.902	0.847	0.792	0.742	5.12	3.39
10	-	0.822	0.773	0.693	0.642	6.82	4.05
10	+	0.928	0.879	0.812	0.763	8.94	6.01
12	-	0.843	0.797	0.721	0.672	7.92	6.21
12	+	0.953	0.909	0.833	0.790	10.13	9.33
14	-	0.846	0.802	0.724	0.676	10.77	8.43
14	+	0.956	0.911	0.837	0.794	13.68	12.09
17	-	0.851	0.805	0.729	0.681	15.91	12.11
17	+	0.958	0.913	0.840	0.799	20.36	17.98

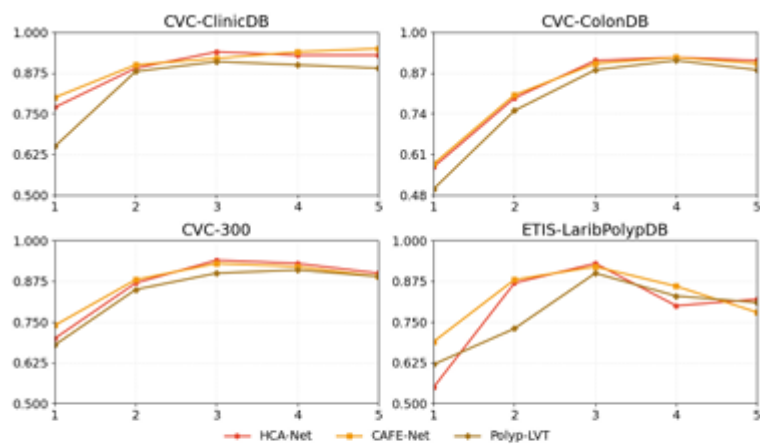


Fig. 6. Performance vs. size on the five polyp datasets. The x-axis denotes the proportion (%) of polyp size, and the y-axis represents the averaged mDice coefficient.

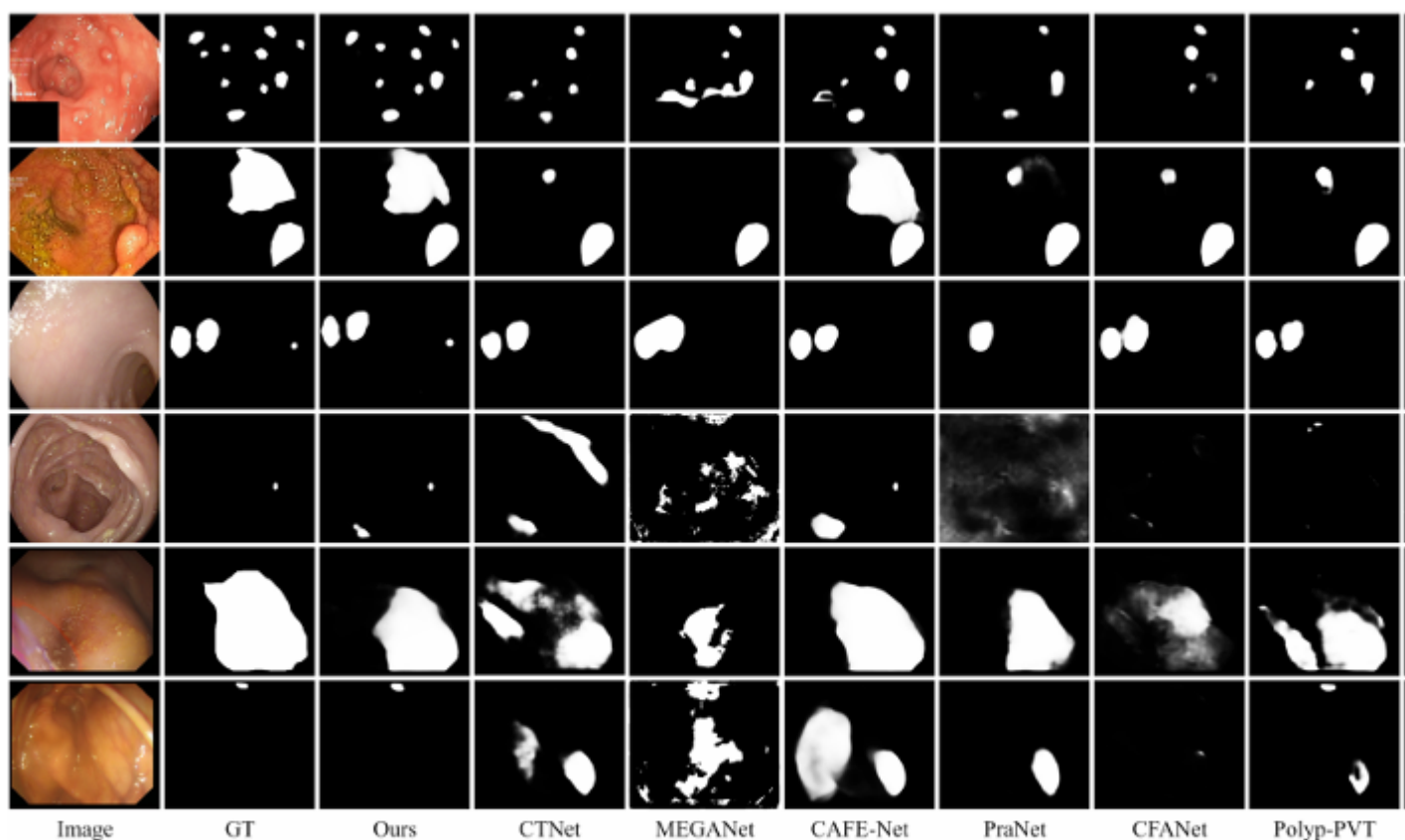


Fig. 4 Our model was compared with the results of various advanced methods on the polyp dataset

- Image(Diagram) of the model with related Modules -> two images
- Heatmap of the model in ablation study with 6 samples
- Heatmap of the model for some stages

- Heatmap of the model in comparison with other models
- Table of the ablation study